Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part – January 17th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:30m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about Parallel Algorithms and Parallel Programming and briefly explain (without an explanation, the answer will be considered invalid)

A. What is meant by *DOALL* loop? Example? (1)

B. Is the following code parallelizable? (1)

```
for (i=10; i<100; i++)
   a[i-10] = a[i] + 100;
```

C. Please describe the differences between flow, output and anti-dependencies. (2)

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the stencil pattern: assumptions, complexity, example. (1)

B. Please describe the differences between Split and Bin parallel patterns. (1)

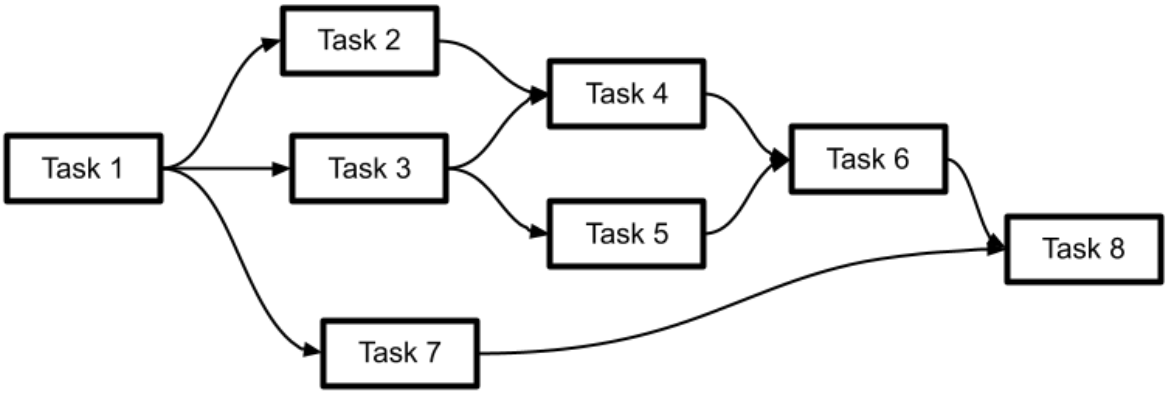
C. Given an input collection of 100 values and an auxiliary array of 10, which parallel pattern between the gather and the scatter is producing an output collection of 10 values? (1)

D. What is meant by *separating hyperplane in a recurrence parallelization*? (1)

Exercise n. 3

Answer the following questions about OpenMP and briefly explain (without an explanation, the answer will be considered invalid)

A. Given an algorithm with the following structure:



And the following values of work: W(Task1) = 2, W(Task2) = 1, W(Task3) = 3, W(Task4) = 2, W(Task5) = 1, W(Task6) = 2, W(Task7) = 5, W(Task8) = 1.

Compute the Work and the Makespan of the algorithm. What is the minimum number of threads required to guarantee the maximum parallelism? (1)

Provide a pseudocode implementation using OpenMP (2)

B. What's the use of the omp critical directive? (1)

Exercise n. 4

A. What are Communicators in MPI? Describe if it is possible to define custom communicators in a program at runtime, and why doing so would be useful. (You can skip this question if you completed the second challenge) (2)

B. What are PThread condition variables and what is their use? (2)